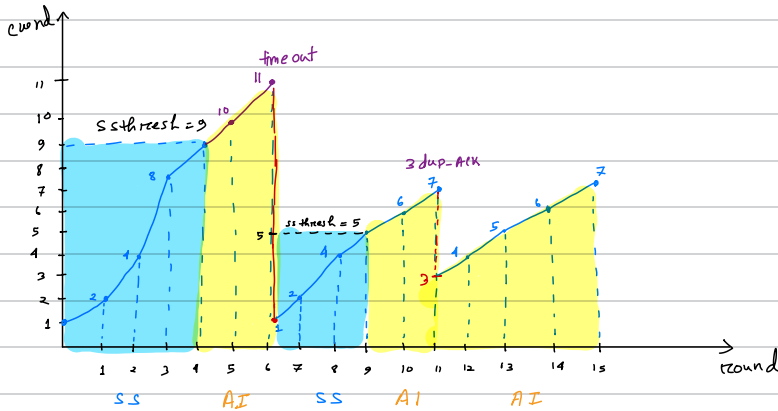


Solution of Congestion Control Math

1. A file transfer starts at $MSS=1$; $ssthresh=9$; $round=0$. The system faces a time-out at around 6 and then a 3-dup-ack on round 11.
 - a. Identify the MSS size on round 15.
 - b. Identify the $ssthresh$ on round 15.
 - c. Total of how many bytes were sent till round 15 if each MSS held 3 bytes of data?



a. MSS size on round 15 is 7.

b. ss threshold will be :

i. For timeout at 6

$$so, 11 \div 2 = \lfloor 5.5 \rfloor = 5$$

ii. For 3 ACK at round 11

$$ssthresh = 7 \div 2 = \lfloor 3.5 \rfloor = 3 \text{ (Ans)}$$

$$\begin{aligned}
 \text{c. Total mss's} &= 1 + 2 + 4 + 8 + 9 + 10 + 11 + 1 + 2 + 4 + 5 \\
 &\quad + 6 + 7 + 3 + 4 + 5 + 6 + 7 \\
 &= 95
 \end{aligned}$$

$$\text{Now, total bytes} = 95 \times 3 = 285 \text{ bytes}$$

2. A file transfer is currently at $MSS = 26$; $ssthresh = 10$, $rtt = 28$

- Identify if there was any congestion along the transmission.
- If the last congestion occurred due to 3 ACKs, what was the value of RTT at that time?
- If the last congestion occurred due to time-out, what was the value of RTT at that time?

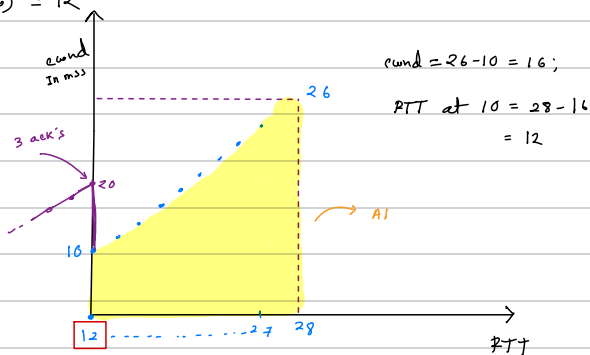
a. Here mss is greater equal than $ssthresh$.

So, there will be congestion along the transmission.

b. $ss\ threshold = 10$, $RTT = 28$, $mss = 26$

If last congestion occurred due to 3 acks, means after that mss increased by 1 each RTT.

so, RTT is $(28 - 16) = 12$

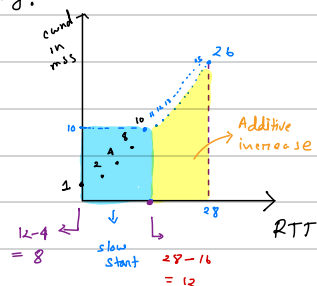


c. If the last congestion occurred due to timeout
 so, After timeout the will be slow start means each round cwnd will increase multiplicatively.

when $cwnd$ is 10, $RTT = 28 - (26 - 10)$
 $= 12$

Before that was slow start.

so, at $cwnd = 1$, $RTT = (12 - 4)$
 $= 8\ (Ans)$

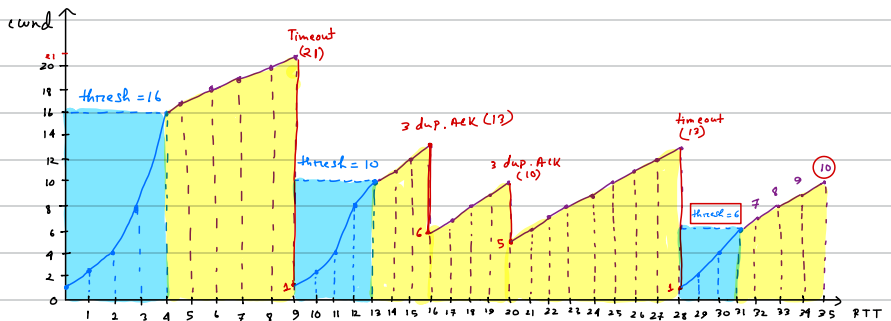


3. In TCP Congestion Control, the sender started (at $RTT=0$) with the slow start and the initial $ssthresh$ was set to 16. Then the following events occurred.

- Event 1: Time-out occurred at $RTT = 9$
- Event 2: 3 Duplicate acks received at $RTT = 16$
- Event 3: 3 Duplicate acks received at $RTT = 20$
- Event 4: Time-out occurred at $RTT = 28$

- Draw the $cwnd$ vs rtt graph for the above scenario.
- Calculate the value of $ssthresh$ at $RTT=35$.
- Calculate the value of $cwnd$ at $RTT=35$.

a.



b. At $RTT 35$, value of $ssthresh$ is 6.

c. At $RTT 35$, value of $cwnd$ is 10.